

Peptic Ulcer Disease

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Outline

- Introduction
- Epidemiology
- Anatomy & physiology
- Etiology of PUD
- Clinical features
- Management

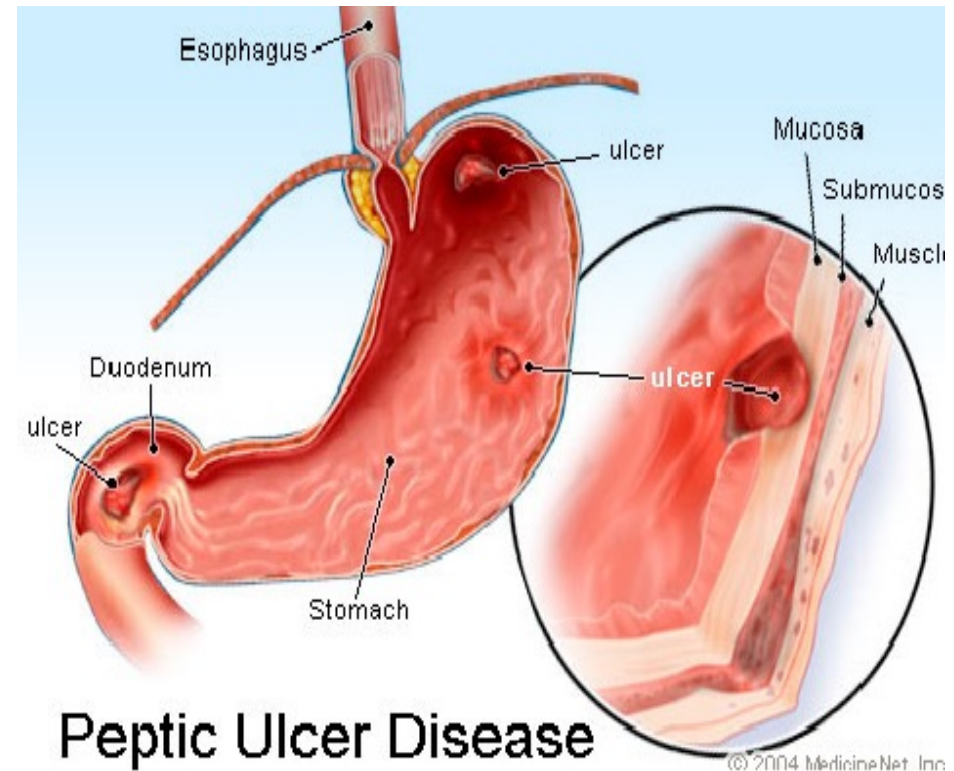
Introduction

- **Ulcer**

- disruption of mucosal integrity (>5mm) leading to *local defect or excavation* due to active inflammation.

- **Peptic ulcer**

- Break in the gastric or duodenal mucosa that penetrates down to muscularis mucosa/submucosa
- PUD encompasses both gastric and duodenal ulcers



Epidemiology

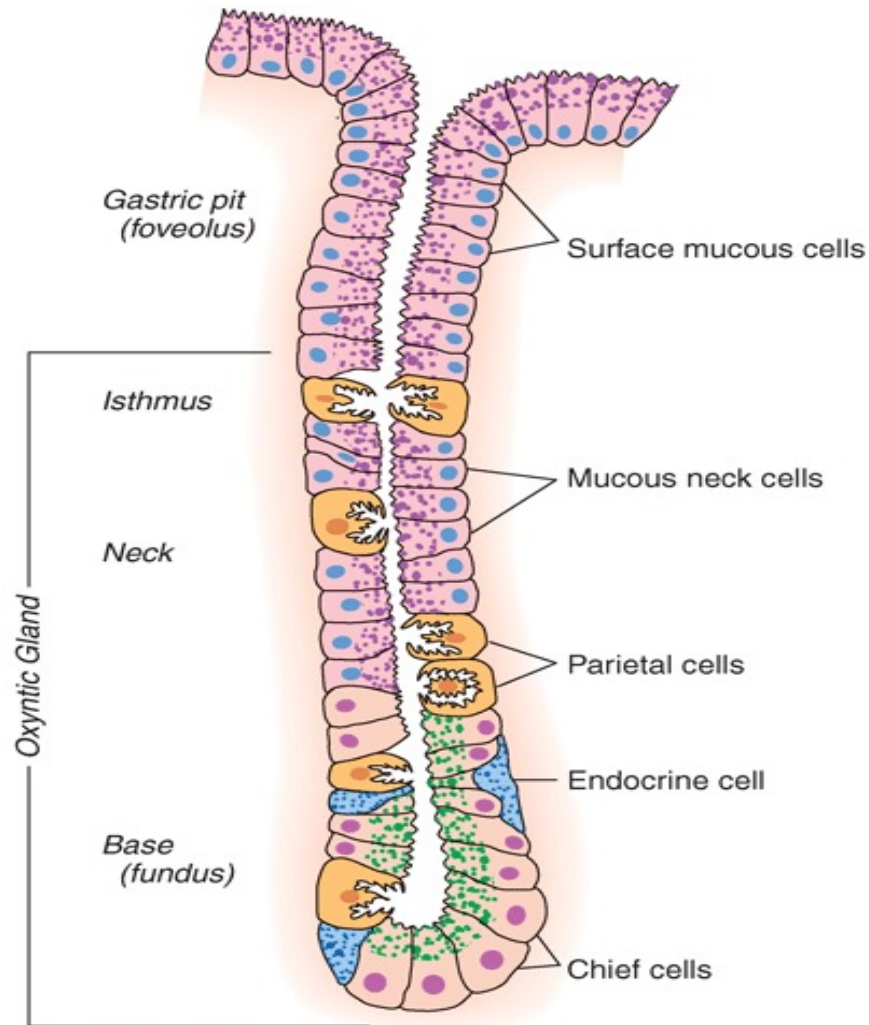
- Rare before 1800s
- M>F (DU 5:2, GU 2:1)
- Helicobacter pylori – 10 fold increase in incidence
- Incidence increases with age
- Prevalence is decreasing
 - Successful therapy of H.pylori
 - Widespread use of PPI/acid suppressive therapy

Pathophysiology

PUD

**A result of imbalance between mucosal
damaging factors and mucosal protective factors**

Gastric Anatomy/Physiology



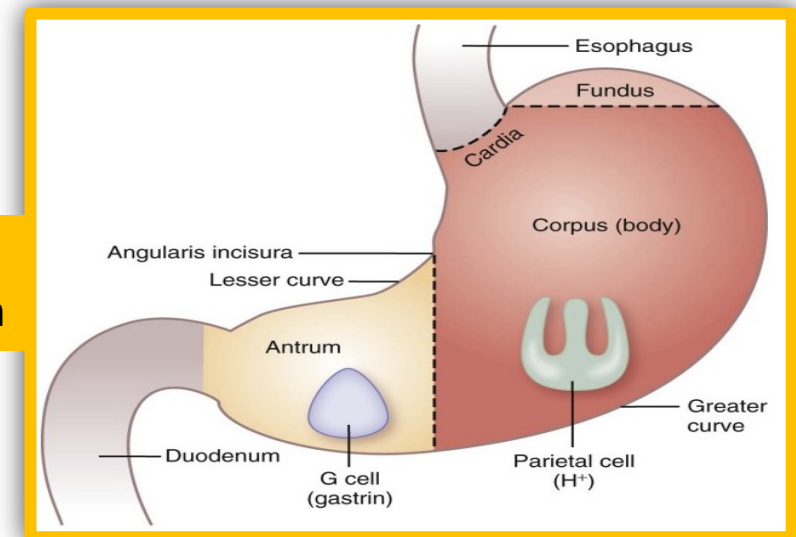
- **Anatomic**

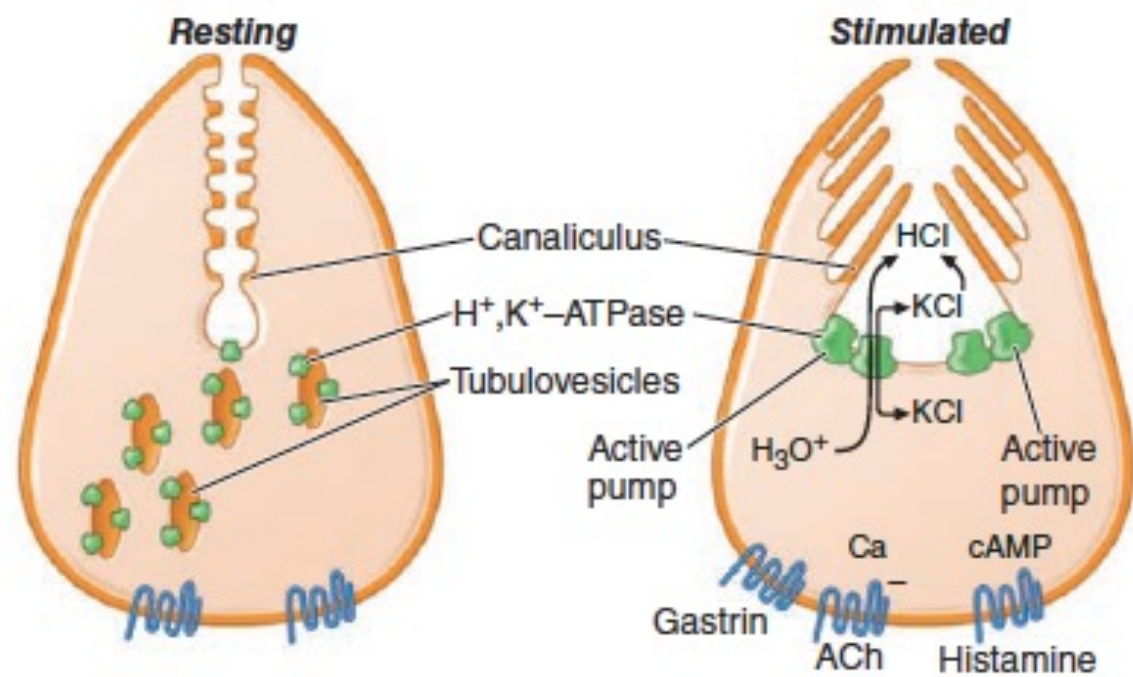
- Fundus, Corpus, Antrum

- **Functional**

- Oxyntic (80%), pyloric gland (20%)

Parietal cells=HCL production
Chief cells= Pepsin production





Stimulants

Histamine:
ECL, mast
cells

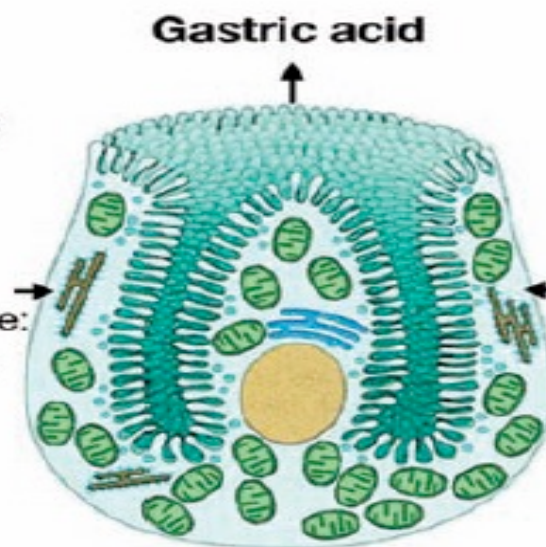
Acetylcholine:
vagus nerve

Gastrin: G
cells

Inhibitors

Somatostatin:
D cells

Prostaglandins:
epithelial and
nonepithelial
cells of the
stomach



Parietal cell

Fig. 3. Physiology of the parietal cell. ECL, enterochromaffin-like.

Gastroduodenal Defense

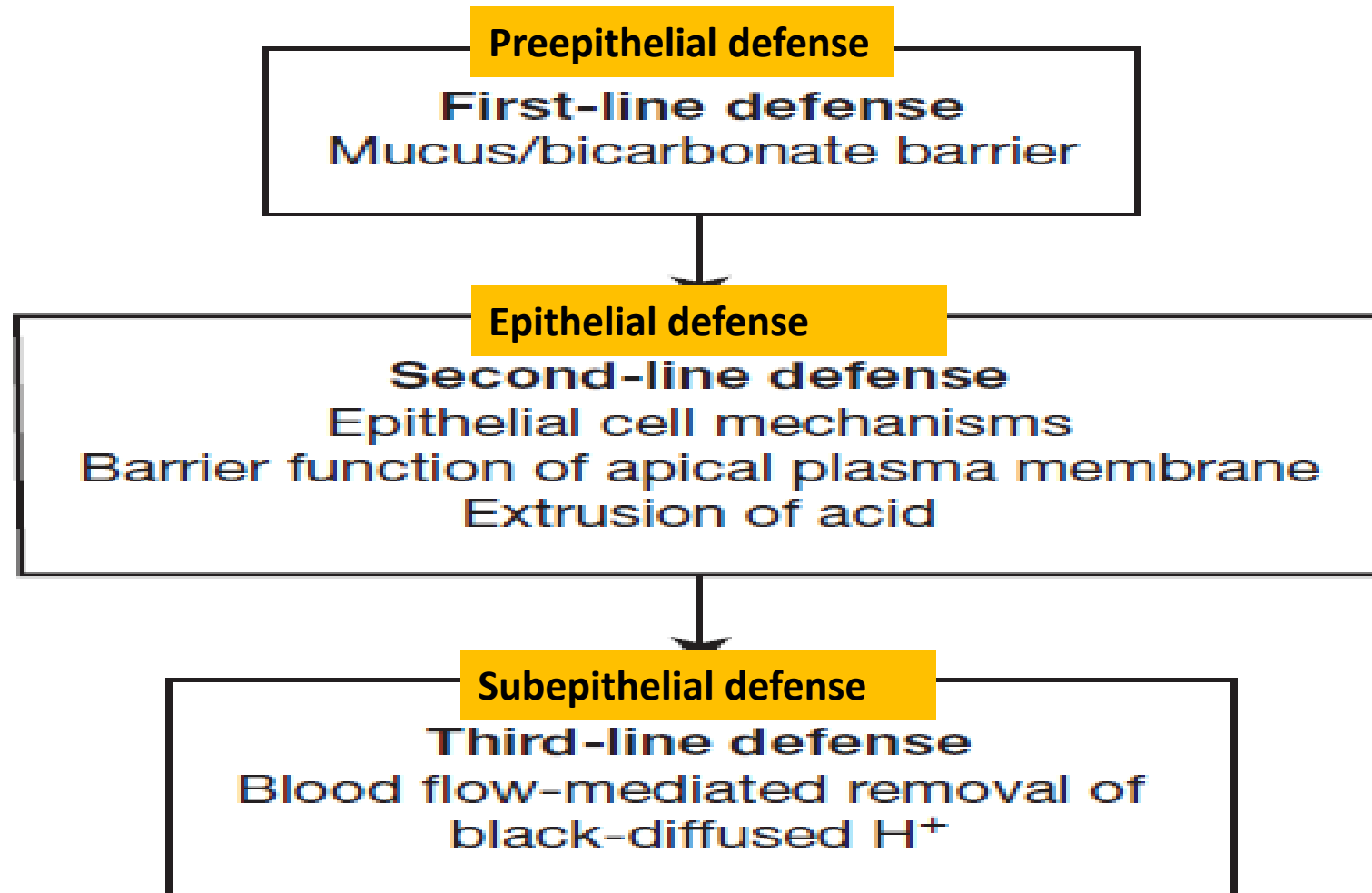
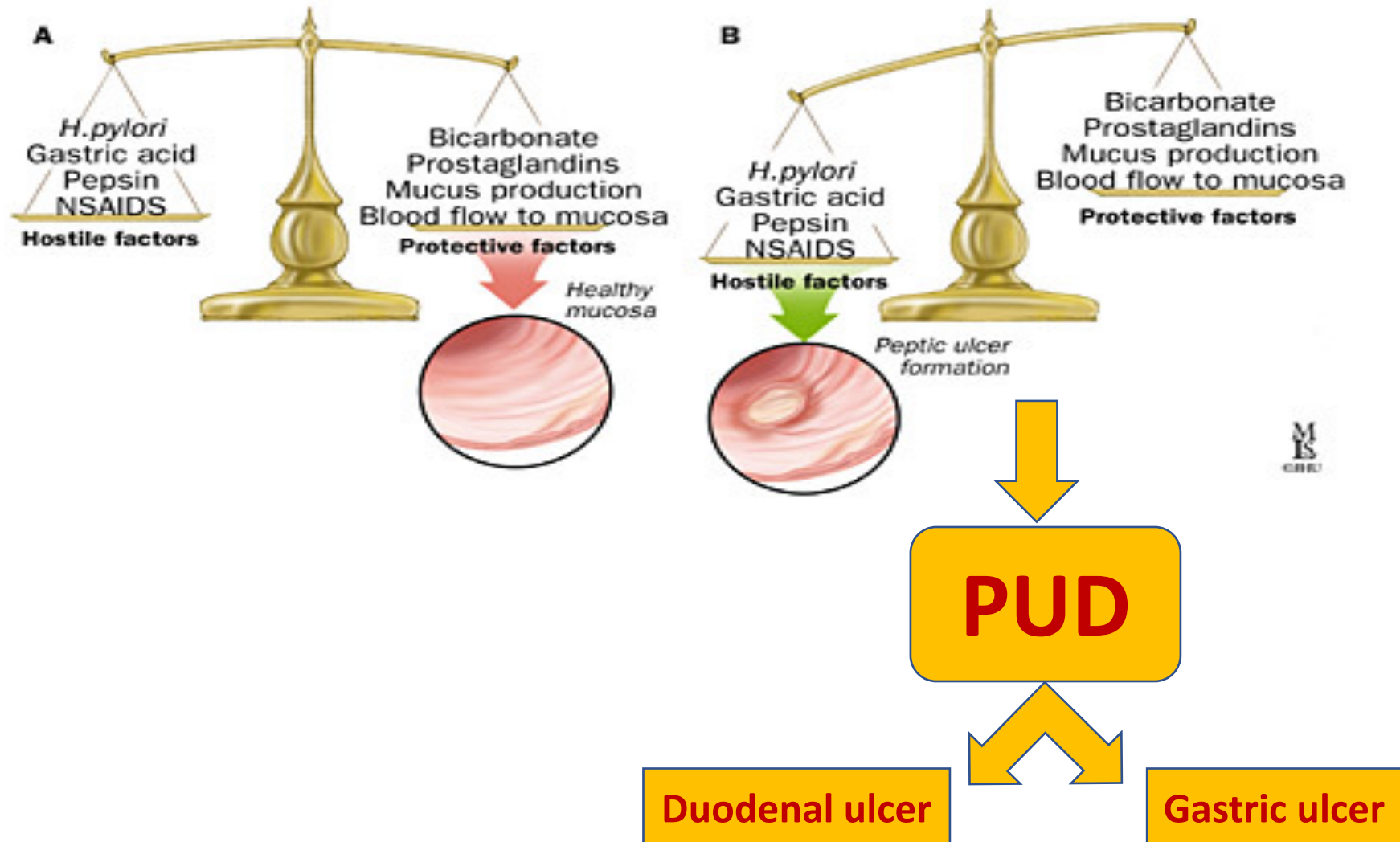


Fig. 2. Mucosal defense mechanisms.

Pathophysiology



Etiology of PUD

- **Helicobacter pylori infection**
- **NSAIDs**
- Others-uncommon
 - Zollinger Ellison syndrome
 - Mastocytosis
 - Crohn's disease

- **Smoking**

- Stimulation of secretory cells
- Risk for PUD in H. Pylori infected
- Delays ulcer healing
- Does not cause ulcer by itself

- **Alcohol**

- Acute Effects
 - Damage mucosal barrier
 - Stimulate acid secretion
- Chronic Effects
 - No evidence that it causes / exacerbates ulcer

- **Diet**

- Essential fatty acids, fruits, vegetables, fiber diet, chili peppers/?protective
- Coffee
 - Stimulate acid secretion, esophageal reflux
 - More for non ulcer dyspepsia
 - No evidence as risk for PUD
- Milk – strong acid secretagogue

- Psychological factors

Clinical Features

History

- **Abdominal pain** is common to many GI disorders
 - *poor predictive value* for presence of either DU or GU.
- **Up to 10%** of pts with *NSAID-induced mucosal disease can present with a complication (bleeding, perforation, & obstruction) without antecedent symptoms.*

- **Epigastric pain** described as *a burning or gnawing discomfort* can be present in both DU & GU.
- The typical pain pattern in DU;
 - Occurs 90 minutes to 3 hours after a meal and
 - Frequently relieved by antacids or food.
- Pain that awakens the patient from sleep (*between midnight & 3 A.M.*) is the most discriminating symptom, with *two-thirds of DU pts describing this complaint*.
 - This symptom is also present in *one-third of pts with NUD*.

- In GU pts, *discomfort may be precipitated by food.*
- **Nausea & weight loss** occur more common in **GU**.
- Endoscopy detects ulcers in **<30%** of pts who have **dyspepsia**.

- Sudden onset of severe, generalized abdominal pain may indicate **perforation**.
- Pain worsening with meals, nausea, and persistent vomiting of undigested food suggests **gastric outlet obstruction**.
- Tarry stools or coffee-ground vomiting indicate **bleeding**.

PUD related complications

GI Bleeding

- Most common complication
- ~15% of patients
- Older age group (>60 yrs)
- 20% - no warning sign & symptoms

Perforation

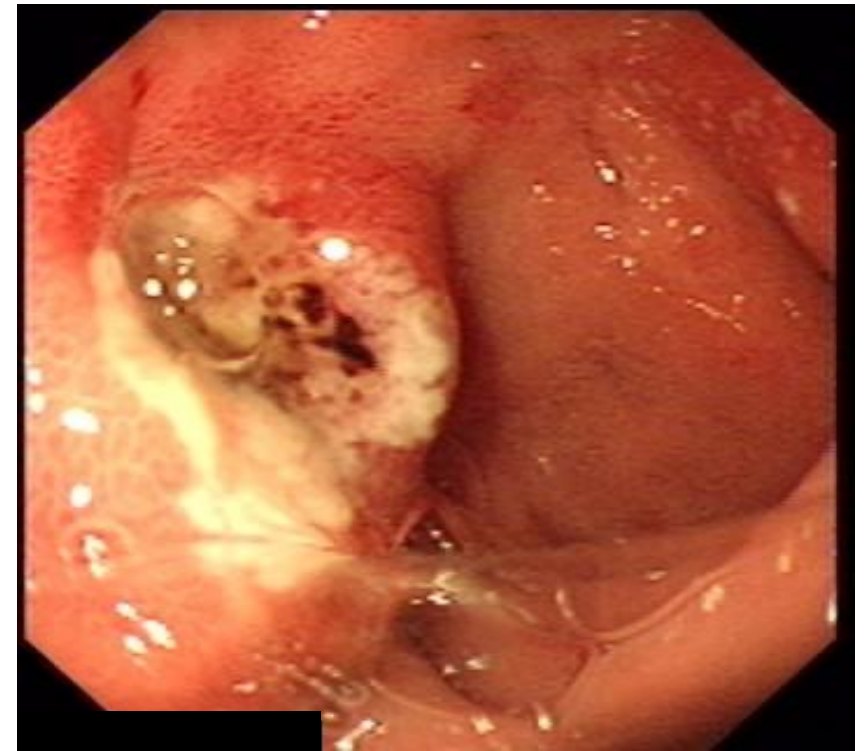
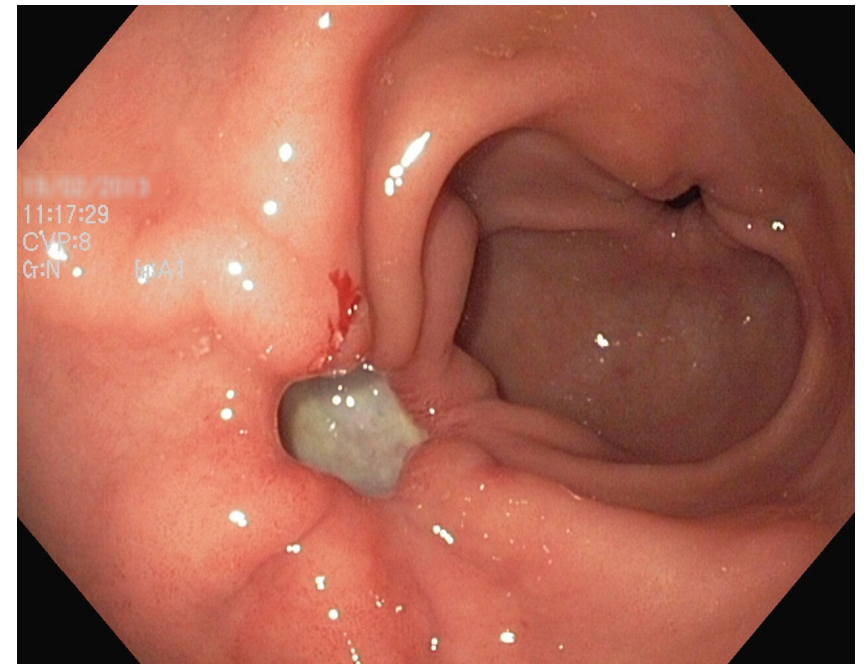
- 6-7% patients with PUD
- ↑ with use of NSAID
- Penetration – to adjacent organ

GOO (Gastric outlet obstruction)

- 1-2% of patients
- Ulcer inflammation or edema vs healed scar (fibrosis)

Endoscopy

- Most accurate diagnostic test for PUD
- Advantages
 - Biopsy – Malignancy & H. Pylori testing
 - Identify and treat source of bleeding
 - Dilation of pyloric/duodenal stenosis



Drugs Used in the Treatment of Peptic Ulcer Disease

Drug Type/Mechanism	Examples	Dose
Acid-suppressing drugs		
Antacids	Mylanta, Maalox, Tums, Gaviscon	100–140 meq/L 1 and 3 h after meals and hs
H ₂ receptor antagonists	Cimetidine	400 mg bid
	Ranitidine	300 mg hs
	Famotidine	40 mg hs
	Nizatidine	300 mg hs
Proton pump inhibitors	Omeprazole	20 mg/d
	Lansoprazole	30 mg/d
	Rabeprazole	20 mg/d
	Pantoprazole	40 mg/d
	Esomeprazole	20 mg/d
Mucosal protective agents		
Sucralfate	Sucralfate	1 g qid
Prostaglandin analogue	Misoprostol	200 µg qid
Bismuth-containing compounds	Bismuth subsalicylate (BSS)	

Abbreviation: hs, at bedtime (*hora somni*).

TABLE 317-5 Recommended First-Line Therapies for *H. pylori* Infection

REGIMEN	DRUGS (DOSES)	DOSING FREQUENCY	DURATION (DAYS)
Clarithromycin triple	PPI (standard or double dose)	BID	14
	Clarithromycin (500mg)		
	Amoxicillin (1 g) or Metronidazole (500 mg TID)		
Bismuth quadruple	PPI (standard dose)	BID	10–14
	Bismuth subcitrate (120–300 mg) or Subsalicylate (300 mg)	QID	
	Tetracycline (500 mg)	QID	
	Metronidazole (250–500 mg)	QID (250) TID to QID (500)	
Concomitant	PPI (standard dose)	BID	10–14
	Clarithromycin (500 mg)		
	Amoxicillin (1 g)		
	Nitroimidazole (500 mg) ^f		
Sequential	PPI (standard dose)	BID	5–7
	PPI, Clarithromycin (500 mg) + Nitroimidazole (500 mg) ^f	BID	5–7
Hybrid	PPI (standard dose) + Amox (1 g)	BID	7
	PPI, Amox, Clarithromycin (500mg), Nitroimidazole (500 mg) ^c	BID	7
Levofloxacin triple	PPI (standard or double dose) + Amox (1 g)	BID	5–7
	Levofloxacin (500 mg)	QD	
	Amox (1 g)	BID	
Levofloxacin sequential	PPI (standard or double dose) + Amox (1 g)	BID	5–7
	PPI, Amox, Levofloxacin (500 mg QD), Nitroimidazole (500 mg) ^c	BID	5–7
LOAD	Levofloxacin (250 mg)	QD	7–10
	PPI (double dose)	QD	
	Nitazoxanide (500 mg)	BID	
	Doxycycline (100 mg)	QD	

Surgical treatment

- Role of surgery in management of PUD is minimal
 - Cure of most peptic ulcers with H.pylori eradication
 - Availability of safe and potent acid suppressive drugs

Indications For Surgery in Peptic Ulcer

Emergency

- Perforation
- Refractory bleeding

Elective

- Complications e.g. GOO
- Recurrent ulcer following gastric surgery